Project 2: Turbojet Engine Design/Off-Design Performance Analysis

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The purpose of this project is to plot the performance three space of a fixed-area missile-class engine with a simple converging nozzle. This will be done by varying the fuel flow rate from 20% to 500% the on design fuel rate, varying the Mach number from 0 to 2.5. Additionally, this will be done at sea level, 4000 m, 8000 m, and 12000 m altitude.

Nomenclature

T = temperature $[K]$

P = pressure $[Pa]$

ρ = density $\left[kg/m^3\right]$

T_t = total temperature $[K]$

P_t = total pressure $[Pa]$

τ = $\frac{T_{t,out}}{T_{t,in}}$

π = $\frac{P_{t,out}}{P_{t,in}}$

η = isentropic efficiency

u = velocity $[m/s]$

M = mach number

Subscripts

0 = freestream

d = diffuser

c = compressor

b = burner

t = turbine

n = nozzle

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I. Introduction

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II. Methodology

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III. Results

A. Sea Level

B. 4000 m

C. 8000 m

D. 12000 m

IV. Conclusion

Appendix